

Po-Sheng Cheng

Master of Industrial Design, Rhode Island School of Design (RISD), Jun. 2026

401-432-5576
bencer@outlook.com
Portfolio: <https://bencer3283.github.io/art/>

B.Sc. Bio-Mechatronics Engineering & B.A Economics,
National Taiwan University (NTU), Jan 2023

Notable Skills

SolidWorks, Fusion, Rhino, Keyshot, Figma, Adobe PS/AI/ID, electronics (Arduino)

Experiences

Product Development Apprentice, Google, Jun. - Sep. 2024

Taipei, Taiwan

- Led a team of 5 to design a Hardware as a Service platform for managing lost items in public facilities.
- Built several mock-ups and prototypes with Figma and Raspberry Pi. Drove the development from a single product to a holistic platform.

Electro-mechanical Engineering Intern, Logitech, Feb. - Jun. 2022

Hsinchu, Taiwan

- Created a brand new breed of keyboard technology by conducting quantitative user research with 50 interviewees and innovating a microcontroller-based prototype.
- Collaborated with product management and design departments to explore marketing opportunities.

College Student Researcher, NTU, Jul. 2021 - Feb. 2022

Taipei, Taiwan

- Built a one-stop, from-acquire-to-analysis solution to improve the user experience of scientific spectral measurement. Mapped existing bottlenecks by being involved in research projects in spectral mapping.
- Awarded the **2021 Technology Innovation Award** by CCMS, NTU and **College Student Research Creativity Award** by National Science and Technology Council of Taiwan with this project.

Projects

MITM Router

- Conceptualized and built a WIFI router that has a disruptive user interaction scheme to raise awareness of the hidden cost of technology.
- Extensive prototyping and user testing of the physical user interface was performed to make critical design decisions.

Haptic Belt

- Designed a custom rigid-flex PCB for a belt that uses novel vibration-based user interface to provide navigation for visually impaired people.
- Performed testing on driver and interfacing ICs and vibration motors using oscilloscope. Implemented STM32 firmware for the vibration motor driver.

PageMate smart bookmark

- Implemented the touchpad-based physical user interface of a smart bookmark that can display reading progress of the user's ebooks.
- Using Figma to create a demo experience that can be updated in real time using my Node.js web service.

(Publication) TeleSHift: Telexisting TUI for Physical Collaboration & Interaction

- Designed and prototyped a 3D tangible user interface that was awarded at 2022 ACM Ubicomp conference.
- Designed the robust interactive mechanisms (SolidWorks) and the microcontroller-based prototype. My design for manufacturability (DFM) improvements also slashed assembling time by 80% during demo.

Po-Sheng Cheng

Portfolio Highlights

Hardware Projects

Full Portfolio Link:

<https://bencer3283.github.io/art/>



MITM Router

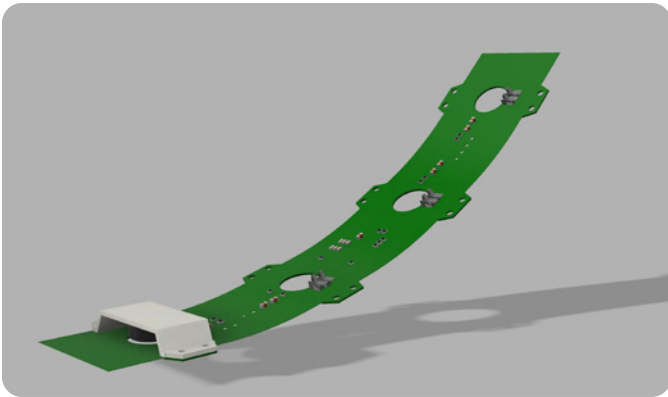
an interactive WI-FI router



Raspberry Pi 5 running Linux networking stack
ESP32 and RP2040 microcontroller
7+ peripheral sensors, actuators or ICs

Haptic Belt

using haptic feedback to provide navigation for visually impaired people



Rigid-flex PCB design with Altium Designer
Power integrity analysis
STM32 firmware programming ([Github Repo](#))
RF circuit design

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ESP32 microcontroller
Extensive use of I2C
Custom IC for touch gesture sensing

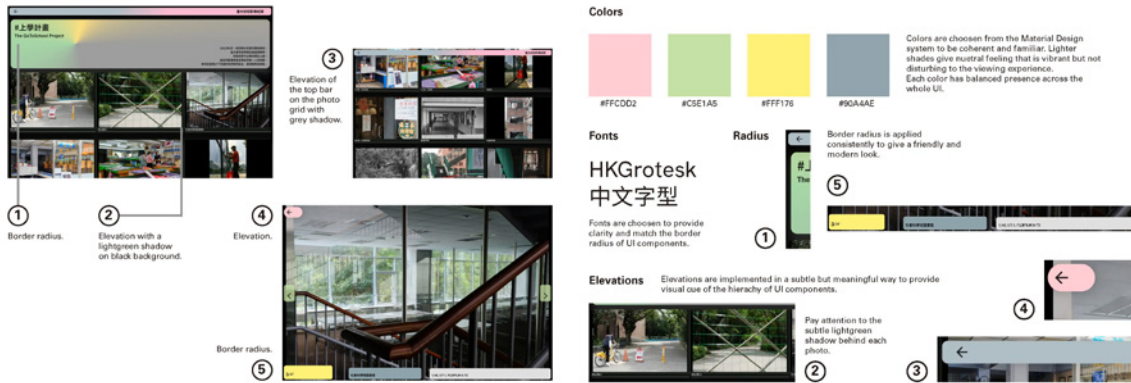
Po-Sheng Cheng

Portfolio Highlights

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My Online Photography Gallery

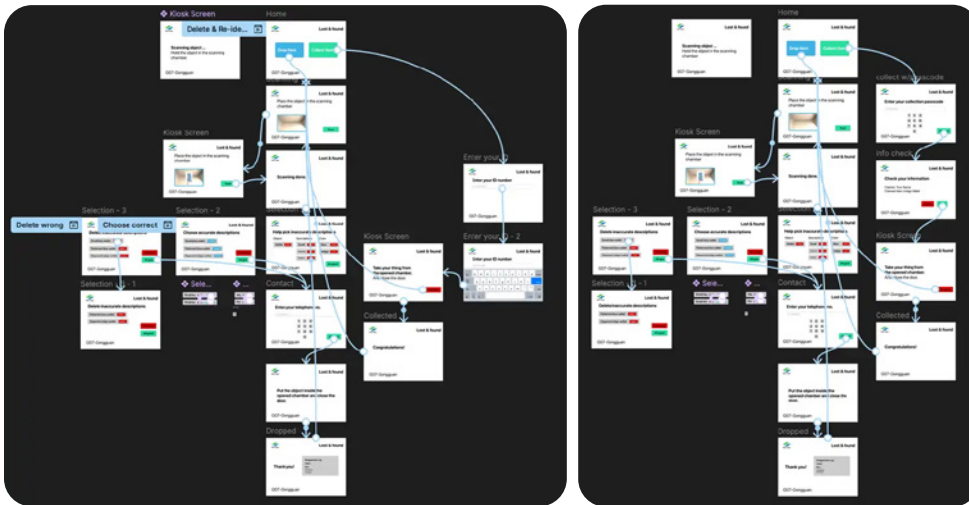


[Design system](#)
[GitHub Repo](#)

Built with: Flutter/Dart, Netlify, Google Drive API

Nexus

intelligent lost item recognition and management platform



Built with: Flutter/Dart, React,
Figma, Linux on Raspberry Pi
[Figma A/B Testing](#)
[GitHub Repo](#)

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